

Ditsa Chakraborty

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PROFILE

Electronics & Communication Engineering student with hands-on experience across machine learning, data analytics, software engineering, and embedded systems. Skilled in Python, SQL, statistical modeling, and full-stack development, with a proven record building and validating ML models (Neural Networks, Random Forest, CNN-based computer vision) at 90%+ accuracy across academic and industry projects. Experience deploying fintech applications and GenAI/NLP pipelines (FinBERT) in a live production environment. Seeking to apply analytical and engineering skills to solve real-world, business-critical problems.

EDUCATION

B.Tech., Electronics & Communication Engineering SRM Institute of Science and Technology Chennai, TN CGPA: 8.33 / 10 (up to 6th Semester)	2023 – 2027
Std XII — CBSE Indus Valley World School Kolkata, WB — 75.2%	2022
Std X — CBSE Indus Valley World School Kolkata, WB — 86%	2020

INDUSTRY EXPERIENCE

InvestTech Analyst Intern ZeTheta Algorithms Pvt. Ltd.	02/2026 – 05/2026
- Deployed and developed four high-quality fintech applications including Market Depth Explorer, Convertible Bond Pricing Framework, Market Monitoring Software, and a Machine Learning-Based Sentiment Analysis Trading Platform. - Built full-stack analytics applications using React, Node.js, Express, TypeScript, and PostgreSQL. - Performed quantitative analysis and statistical modeling using Python and Pandas; created an NLP pipeline using FinBERT to analyze sentiments, generate trading signals, and run simulations. - Evaluated and validated models and pipelines for performance and reliability; visualized KPIs and results using interactive dashboards built on Plotly. - Maintained version control using Git and followed CI/CD deployment processes using Render and Vercel.	

VOLUNTEER EXPERIENCE

Volunteer, My Children Foundation (NGO)	06/2025 – 07/2025
Contributed to digital outreach initiatives supporting underprivileged children; edited and transcribed YouTube videos and created short-form content for social media and fundraising campaigns.	

ACADEMIC PROJECTS

Transmission and Reception of Signal Using ESP-32 with Machine Learning SRM Institute (Team)	05/2026
- Built and deployed a multi-nodal communication system using ESP32 devices for signal transmission and reception experiments. - Conducted performance experiments to gather and analyze RSSI data under various conditions and beams. - Used Neural Network and Random Forest classification algorithms to predict communication behavior with 99.7% and 90.65% accuracy, respectively. - Built a real-time Python dashboard for tracking communication data. - Presented work at RASET 2026 – a National Conference organized by IEEE OES.	
Fruit Quality Assessment using MobileNetV2 SRM Institute	04/2026
- Designed and implemented a Convolutional Neural Network (CNN) machine vision pipeline using MobileNetV2 with transfer learning for classifying fresh vs. spoiled fruit, achieving 96.1% accuracy. - Conducted data pre-processing, model training, testing, and validation. - Completed technical documentation and material for an IEEE publication.	
Light-Activated Smart Fan SRM Institute (Academic Paper)	11/2025

- Designed and developed an automated control system using Arduino Uno, LDR sensors, and relay modules.
- Conducted testing, troubleshooting, and validation under varying operating conditions.
- Documented methodology and findings in an academic paper submitted at SRM (Turnitin similarity: 8%).

TECHNICAL SKILLS

Data Analytics & Machine Learning: Pandas, Scikit-Learn, Data Analysis, Data Visualization, Plotly, Neural Networks, Random Forest, MobileNetV2, FinBERT, NLP

Embedded Systems & ECE: ESP32, Arduino, FPGA/Verilog, IoT, LTspice, Scilab

Programming Languages: Python, SQL, Java, TypeScript, JavaScript

Databases & Cloud: PostgreSQL, Supabase, Vercel, Render

Software Development: React, Node.js, Express.js, REST APIs, Git, CI/CD

Finance / Quant: Financial Modeling, Quantitative Analysis, Market Microstructure, Algorithmic Detection

Professional Skills: Research, Technical Documentation, Problem Solving, Presentation Skills, Team Collaboration, Design Thinking

CERTIFICATIONS & COURSES

Claude 101, Anthropic	07/2026
AI Fluency: Framework & Foundations, Anthropic	07/2026
Certificate of Presentation — RASET 2026 (IEEE OES), Bannari Amman Institute of Technology	05/2026
Python for Data Science, NPTEL – IIT Madras	07/2024 – 08/2024
Certificate of Project Work — Sentiment Analysis Trading, ZeTheta Algorithms	05/2026
AI Foundations for Everyone, Coursera – IBM	04/2025
Design Thinking – A Primer, NPTEL – IIT Madras	01/2025 – 02/2025
Renewable Energy & Green Building Entrepreneurship, Coursera – Duke University	04/2026
Certificate of Project Work — Phantom Liquidity Hunter, ZeTheta Algorithms	03/2026
Certificate of Project Work — Convertible Bond Laboratory, ZeTheta Algorithms	03/2026
Certificate of Project Work — Market Depth Explorer, ZeTheta Algorithms	04/2026
Learning Verilog for FPGA Development, LinkedIn Learning	11/2025
Unlocking IoT Potential with Arduino (Workshop), SRM Institute	09/2023

ACHIEVEMENTS

Member, Student Council; Organiser, IVWS School Fest – Art	2019 – 2020
• Member, Student Council (2020). • Organiser, IVWS School Fest – Art (2019).	
First Prize — Cooking Without Fire, 15th Annual Cultural Competition (2024)	